



OPEN Harris Hawks–tuned severity-aware YOLOv8 instance segmentation framework for vehicle damage assessment

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The severity of vehicle damage is very important in determining road safety, insurance claim and automated vehicle inspection. Although the recent methods of computer vision were promising in the localization of damaged areas, it is difficult to discriminate the level of damage precisely because of subtle visual variations, class disparity, and irregular damage images. In a bid to solve these problems, this paper suggests a severity-sensitive YOLOv8-based instance segmentation system, which incorporates curriculum learning, Harris Hawks Optimization (HHO) to hyperparameter optimization, and confidence thresholding at the per-class level. The proposed learning strategy in the curriculum allows the training to be progressive, whereby initial training occurs with understanding of types of coarse damage, and then further training is given in fine-grained severity discrimination. HHO is used to automatically tune important hyperparameters with severity-sensitive fitness objectives to enhance the robustness of segmentation without changing the architecture. Also, class-specific confidence thresholds are proposed to adjust the precision and recalling to the levels of severity. Experimental analysis of a curated vehicle damage severity dataset shows that convergence is stable and the performance is competitive, with Box mAP50 of 0.271 and Mask mAP50 of 0.135, showing the best performance at an early training phase. The proposed lightweight YOLOv8s-Seg model has shown to be better than a larger YOLOv8m-Seg baseline with practical thresholds of IoU, which proves its applicability in practice. The confusion matrix and class-wise analysis indicate the presence of a strong detection of severe damages, whereas surface defects that emerge in subtle cases are noted as a challenge. In general, the proposed framework offers an effective, understandable, and implementable algorithm of automated damage severity in vehicles.

Keywords Vehicle damage assessment, Severity-aware learning, YOLOv8 instance segmentation, Curriculum learning, Harris Hawks Optimization, Hyperparameter optimization, Per-class confidence thresholding

The assessment of the severity of vehicle damage has become a significant part of the contemporary transportation system, especially in the road safety, insurance claims and fleet management, and automated vehicle inspection. Conventionally, evaluation of damage has been extremely dependent on manual inspection by professionals, which is time consuming, subjective and in most cases inconsistent between inspectors. As the number of vehicles and traffic density grows at an alarming rate, the necessity to find automated and reliable solutions that will be able to detect the presence of damage and the level of damage of the slightest, moderate, or severe impact progressively increases¹.

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The recent progress in computer vision made it possible to create image-based systems that can analyze the damage on vehicles directly based on visual information. Using the inspections or surveillance high-resolution images, these systems are able to localize areas of damages and distinguish between defect types (dents, scratches, and broken parts). The classification of severity is an important part of the process because it directly determines the estimation of repair costs, insurance choice, and safety evaluation. Fine-grained severity recognition enables the stakeholders to set priorities on the repairs, minimize fraudulence claims, and enhance transparency on the post-accident assessment^{2,3}.

Although there have been enormous improvements, the vehicle damage severity analysis is a daunting task as a result of subjective differences in light conditions, models of different vehicles, degree of damage, and clutters of the background. Less serious damages like scratches can be physically invisible whereas serious damages can have complicated forms and uneven edges. To overcome these issues, there is a need to have strong datasets and powerful vision models that are able to localize and segment accurately. Consequently, studies in the area of vehicle damage severity detection have been steadily on the increase in order to provide valid, scaling, and real-time solutions to intelligent transportation and automotive industry uses^{4–6}.

Problem formulation

Vehicle damage severity assessment can be mathematically formulated as a multi-class instance segmentation and severity classification problem defined over a set of vehicle images. Let the dataset be represented as

$$D = \{(I_i, A_i)\}_{i=1}^N \quad (1)$$

where $(I_i \in \mathbb{R}^{H \times W \times 3})$ denotes the i -th RGB vehicle image of height H and width W , and A_i represents the set of K annotated damage instances in that image. Each instance is defined by a binary segmentation mask M_{ij} and an associated severity class label c_{ij} , where C corresponds to the predefined damage–severity categories. The objective of the learning model is to estimate a mapping function

$$f_\theta = I_i \rightarrow \{(B_{ij}, M_{ij}, c_{ij})\}_{j=1}^K \quad (2)$$

parameterized by θ , which predicts for each detected damage instance a bounding box B_{ij} , a segmentation mask M_{ij} , and a severity class c_{ij} . The bounding box $B_{ij} = (x_{ij}, y_{ij}, w_{ij}, h_{ij})$ localizes the damaged region, while the mask M_{ij} provides pixel-level delineation of the damage. Severity prediction assigns each instance to one of the predefined levels such as minor, moderate, or severe^{7–10}.

Model optimization is carried out by minimizing a composite loss function that jointly accounts for localization, segmentation, and severity classification errors:

$$L = \lambda_1 L_{box} + \lambda_2 L_{mask} + \lambda_3 L_{cls} \quad (3)$$

where L_{box} measures bounding box regression error (e.g., IoU-based loss), L_{mask} quantifies pixel-wise discrepancy between predicted and ground-truth masks, and L_{cls} represents multi-class severity classification loss, commonly defined using categorical cross-entropy:

$$L_{cls} = - \sum_{c=1}^C y_c \log(p_c) \quad (4)$$

with y_c denoting the ground-truth severity label and p_c the predicted class probability. The optimization seeks parameter values θ^* that minimize L , thereby enabling accurate localization, segmentation, and severity-level discrimination of vehicle damage instances across diverse visual conditions. There are following research contributions for this work:

1. Proposed a severity-aware curriculum YOLOv8-Seg, achieving Box mAP@50=0.271 and Mask mAP@50=0.135 for vehicle damage assessment.
2. Applied Harris Hawks Optimization for hyperparameter tuning, improving fine-grained segmentation with Mask mAP@50–95=0.045.
3. Proposed per-class confidence thresholding, yielding high prediction reliability with precision up to 0.97 for severe damage classes.
4. Validated a lightweight YOLOv8s-Seg model, outperforming YOLOv8m-Seg at practical IoU thresholds (0.271 vs. 0.203 Box mAP@50).
5. Demonstrated early optimal convergence, with peak performance at epoch 22, enabling efficient early-stopping deployment.

The rest of this paper will be structured in the following way. In Section II, the proposed severity-aware YOLOv8 instance segmentation framework, curriculum learning, and hyperparameter optimization are discussed. Part III is the description of the experiment, the information about the dataset and the measures of evaluation. The quantitative and qualitative findings are addressed under Section IV and the conclusion and future research directions are presented under Section V.

Review of literature

The end-to-end Mallios et al.¹¹ damages severity estimation pipeline estimate pipeline was proposed using real-world insurance claim images that were taken through mobile phones. Semantic segmentation was used to

determine the damaged vehicle panels and extracted visual features were combined with structured claim data using XGBoost. A large historical claims dataset was experimented with to demonstrate better severity and cost estimation accuracy than structured-data-only baselines, allowing claims to be processed more quickly.

Lin and Chen¹² came up with a LiDAR-based system of vehicle damage assessment with smartphone-collected point clouds. ICP with deep-learning refinement correlation was used to align pre- and post-collision 3D vehicle models. The technique minimized RMSE to 0.29 units and rotation error to 0.04°, which is high in terms of quantification of damage.

Pose et al.¹³ introduced a data-driven propeller damages recognition and the extent of damage estimation with real UAV flight inertial measurements. Three levels of damage were added and neural networks with classifiers were used to determine the type of damage, its severity and the damaged rotor. Experiments on flights proved that localization of faults can be performed correctly without visual sensors.

Agrawal et al.¹⁴ study suggests a hybrid framework consisting of a YOLOv8-based automated vehicle damage detector and segmentation that can be used in insurance. The model combines the task of object detection with semantic segmentation to detect 21 car parts and 8 types of damages based on a dataset of more than 4000 images. The system is better in terms of accuracy and processing speed than the traditional CNN-based methods, and thus it is applicable in real-time assessment of claims. The findings prove improved accuracy in the detection of the level and location of damage. Nevertheless, it has shortcomings such as reliance on the quality of data sets and lack of robustness in other lighting and real-life conditions.

The article by Loureiro et al.¹⁵ is a benchmarking analysis of various deep learning models to detect and segment vehicle damage in shared mobility systems. Models such as Mask-DETR and YOLOv8 performed well using a dataset of 1812 images that had 21 vehicle parts and 8 types of damage, wherein YOLOv8 had a high performance of about 88.9% mAP50. The paper points out the tradeoff between speed and accuracy of inference between models. Findings affirm that the transformer-based architecture is more effective than the conventional CNNs on segmentation tasks. Limitations, however, comprise imbalance in the datasets and absence of a wide variety of real-world damage cases.

Khalid et al.¹⁶ suggest a saliency-based segmentation method with a custom AE-Net architecture to fine-grained vehicle damage detection. The model becomes trained on the CarDD dataset of 4000 images and more than 9000 annotated instances of damage. The method enhances pixel-level segmentation especially on small and irregular damages such as cracks and scratches. The experimental performance demonstrates a high level of performance relative to YOLO and Mask R-CNN in the detection of minor damage patterns. The method, however, is computationally-intensive and has difficulty with very low-contrast regions of damage.

Lokeswari et al.¹⁷ introduce a Mask R-CNN framework of automatic vehicle damage detection and cost estimation of repairs. The model reads and extracts features (type of damage, size, and location) of the image to estimate the cost of the repair process in a systematic manner. The system has an accuracy of 88.78% and an F1-score of 0.880, indicating its effectiveness in segmentation and cost prediction. Findings show enhanced consistency and less human bias in insurance analysis. Nonetheless, variability of data in the set and difficulties of precise estimation of cost of complex damages restrict the performance of the model.

Aithal et al.¹⁸ introduce a Vehicle Damage Assessment System (VDAS) which combines YOLOv5 and Mask R-CNN to detect and estimate costs automatically. The system transforms high-resolution images of vehicles and identifies the types of damage (dent, scratch, etc.) with trained models. YOLOv5 model has an F1-score of 0.39 which has an accuracy of 71.9% and allows real-time assessment in a GUI interface. The method enhances efficiency and minimizes the use of manual inspection. Nonetheless, it has drawbacks such as moderate precision and reliance on predetermined cost limit.

Reddy et al.¹⁹ introduce a deep learning system that includes CNN and R-CNN models to detect and classify vehicle damage. The system recognizes where, what, and how much damage has been done based on user-produced photos and helps to estimate insurance claims automatically. The method focuses on real-time processing and better classification accuracy by the methods of feature extraction and optimization. Findings indicate an excellent automation and less human intervention in damage assessment. Nonetheless, drawbacks are a lack of training data and difficulties with managing varied types of damages.

Suriyapathiraja et al.²⁰ present a Generative AI-oriented system that uses GPT-style models to detect vehicle damage and estimate the cost. The framework is based on multimodal inputs and rule-based logic to recognize the types of damage and approximate cost of repair with high precision. It saves a lot of time processing and improves decision-making as opposed to the traditional methods. Findings confirm its usefulness as compared to expert appraisal in actual situations. But there are shortcomings such as reliance on timely design and possible errors in the interpretation of complex damages.

Chaubal et al.²¹ suggest a vehicle damage detection system that is based on deep learning and transfer learning on ResNet-50 and EfficientNet-B0 models. The system refers to the TQVCD to categorize damage according to car orientation and the severity. Findings indicate that there is high classification accuracy and better generalization in different conditions. The model saves on time of manual inspection and increases the efficiency of claim processing. Limitations are however, reliance on small labeled datasets and difficulties with real-time deployment.

Liu and Wu²² introduce DEH-YOLO, the improved version of YOLOv8 that with the help of deformable convolution and multi-scale attention, the model will effectively address the problem of object misalignment. The model will enhance the identification of minor and anomalous damages of vehicles in the presence of a tailor-made dataset. Experimental results show higher accuracy and efficiency compared to baseline YOLO models. The strategy enhances the performance of detection in challenging backgrounds and in real-world scenarios. Limitations however, consist of added model complexity and computation costs.

Kruacharee et al.²³ introduce DentAnalyzer that is a hybrid deep learning system based on real-life and artificial datasets to evaluate vehicle damage. EfficientNetB5 is applied to this model to identify 15 car parts and

determine the level of damage among four. Findings indicate that the classification of damage and estimation of parts are 94 and 78% accurate, respectively, indicating that there is a high degree of generalization. The use of synthetic data enhances robustness and solves the problem of data scarcity. Such constraints, however, are the dependence on synthetic data quality, and the possible domain drift between simulated and real-world images.

The study by Jalili et al.²⁴ suggests a deep learning-based EfficientNet-based vehicle damage insurance check to automate the claim validation and fraud detection. The model is trained using a dataset of various types of damage, e.g., crack, dent, scratch, and glass shatter collected based on actual insurance claim records. According to experimental findings, EfficientNet performs better than other models, such as ResNet, VGG16, and AlexNet, in two aspects: they exhibit less validation loss and achieve greater classification accuracy. The system will save a lot of claim processing time and improve on reliability in decision-making. Nonetheless, there are weaknesses such as imbalance in dataset and poor performance on minority damage classes, which influence generalization.

Matam et al.²⁵ introduce a hybrid model that integrates YOLOv8 to detect the damage and XGBoost to estimate the cost of repair in a PySpark big data infrastructure. The system is tested on the Kaggle Car Damage Detection dataset which has the following classes scratches, dents and cracks. The outcomes are good with a precision of 91.4, a recall of over 92.0 and mAP50 of 0.70, surpassing the performance of YOLOv5 and CNN baselines. The integrated pipeline will allow efficient and scalable real-time damage estimation and cost prediction. Limitations however are reliance on structured metadata to estimate costs and difficulties in managing very complex or invisible patterns of damage. Table 1 depicts the summary of state-of-the-art studies on damage severity assessment and indirect damage detection across vehicles and infrastructure systems.

Research gaps

The body of research on damage severity assessment highlights several important research gaps. One of the key weaknesses is that it is highly dependent on data quality and modality. Vision-based methods like Agrawal et al.¹⁴, Loureiro et al.¹⁵, and Jalili et al.²⁴ are extremely sensitive to image quality, light, and imbalance in datasets and lack robustness in the real world. In the same way, techniques by Mallios et al.¹¹ and Matam et al.²⁵ are based on structured metadata, whereas LiDAR-based protocols such as Lin and Chen¹² need costly sensors and references of pre-damage, which limits their use in practice and scalability.

The other significant gap is the poor management of small and little damages and absence of cohesive frameworks. Whereas Khalid et al.¹⁶ enhance pixel-level detection, their algorithm is computationally intensive and Liu and Wu²² add complexity to their model to detect smaller objects. Subtle damage detection remains an issue in many models such as Reddy et al.¹⁹ and Aithal et al.¹⁸. Besides, the current systems like Lokeswari et al.¹⁷, Suriyapathiraja et al.²⁰ are trying to estimate costs but do not have completely combined, generalizable end-to-end pipelines.

No.	Authors	Application	Method	Key results	Limitations
1	Mallios et al. ¹¹	Vehicle damage severity	Segmentation + XGBoost fusion	Higher severity and cost estimation accuracy than structured-only models	Dependent on image quality and claim metadata availability
2	Lin and Chen ¹²	Vehicle collision damage	LiDAR + ICP + DL	RMSE 1.27 → 0.29, rotation error 4.03° → 0.04°	Requires LiDAR sensing and pre-damage reference models
3	Pose et al. ¹³	UAV propeller damage	Inertial-sensor ML	Accurate fault localization and severity estimation	Tested on limited damage scenarios
4	Agrawal et al. ¹⁴	Vehicle damage detection & insurance claims	YOLOv8 + semantic segmentation	High accuracy in detecting 21 parts & 8 damage types; improved speed	Sensitive to dataset quality and real-world variations
5	Loureiro et al. ¹⁵	Shared vehicle damage assessment	Benchmarking (YOLOv8, Mask-DINO, RT-DETR)	YOLOv8 ~ 88.9% mAP50; Mask-DINO 88.6% segmentation performance	Dataset imbalance and limited real-world diversity
6	Khalid et al. ¹⁶	Fine-grained damage segmentation	AE-Net (Saliency-based U2-Net)	Superior pixel-level detection for small damages	High computational cost; low-contrast detection issues
7	Lokeswari et al. ¹⁷	Damage detection & cost estimation	Mask R-CNN + cost modeling	88.78% accuracy; F1-score 0.880	Limited performance on complex damage scenarios
8	Aithal et al. ¹⁸	Automated damage assessment (VDAS)	YOLOv5 + Mask R-CNN	71.9% accuracy; real-time GUI-based system	Moderate accuracy; threshold-dependent cost estimation
9	Reddy et al. ¹⁹	Damage detection & classification	CNN + R-CNN hybrid	Improved automation and classification efficiency	Limited dataset size; poor generalization
10	Suriyapathiraja et al. ²⁰	Damage detection & cost estimation	Generative AI (GPT-based) + rule-based system	High accuracy and reduced processing time	Prompt dependency; complex damage misinterpretation
11	Chaubal et al. ²¹	Insurance claim damage classification	ResNet-50 + EfficientNet (transfer learning)	High classification accuracy; robust under variations	Limited labeled data; real-time deployment challenges
12	Liu and Wu ²²	Vehicle damage detection	DEH-YOLO (improved YOLOv8 + attention)	Improved small object detection accuracy	Increased computational complexity
13	Kruacharee et al. ²³	Damage severity & part estimation	EfficientNetB5 + synthetic + real data	94% classification accuracy; improved generalization	Synthetic-real domain gap
14	Jalili et al. ²⁴	Insurance fraud detection & verification	EfficientNet	Superior accuracy over ResNet, VGG16	Dataset imbalance affects minority classes
15	Matam et al. ²⁵	Damage detection + cost prediction	YOLOv8 + XGBoost + PySpark	Precision 91.4%, mAP50 = 0.70; scalable system	Requires metadata; struggles with unseen damage

Table 1. Summary of state-of-the-art studies on damage severity assessment.

Lastly, there are issues of data paucity, generalization and implementation, which have not been addressed. Synthetic data are used in works such as Kruacharee et al.²³, but this has been found to have domain gaps, whereas Chaubal et al.²¹ and Reddy et al.¹⁹ note the lack of labeled datasets. Also, most of the models that perform well and are computationally expensive, limiting real-time and edge deployments. These shortcomings underscore the necessity to have a lightweight, multi-modal and scalable framework that can be used to estimate the severity and cost of various types of damage in the real world under a wide range of conditions.

Materials and methods

Dataset

The damage-severity computer vision dataset is well selected to conduct instance segmentation tasks that can be used to identify and localize various degrees of physical damage. The data has a total of 1,820 images with an instance level annotation to accurately define regions of damages. It has eight damage classes with a wide spectrum of severity: minor dent, minor scratch, moderate broken, moderate dent, moderate scratch, severe broken, severe dent and severe scratch. The class design allows the fine grained learning of type of damage, as well as its severity, both of which are essential to automated inspection and assessment applications.

Figure 1 illustrates a representative case of severe structural damage, characterized by fractured components and extensive material loss, used to motivate severity-aware damage segmentation in this study. In order to have a sound model training and evaluation, the data is carefully split into training, validation and testing sets. The training set has 1,600 images (88%), which offers a sufficient amount of diversity to learn complicated spatial and contextual patterns. The validation set has 144 images (8%), which are useful in hyperparameter optimization and early performance, whereas the test set has 76 (4%) images that are reserved only to unbiased final evaluation. This division ensures that the portion of types of damage in subsets is balanced so that the performance evaluation is consistent.

Each image in the data is preprocessed in a standardized way so that consistency is upheld in training. All pictures are scaled by stretching to a resolution of 640×640 pixels, allowing them to be compatible with the current deep learning framework and allowing the important visual aspects to be retained. Data augmentation is also used to improve generalization and the three variations of outputs are obtained in accordance with each training image. Horizontal flipping and Gaussian blur with an intensity of up to 1.5 pixels are augmentation techniques designed to teach the model invariance to rotation effects and small-scale visual perturbations that are typical of damage that happens in real-world applications.

Preprocessing

To achieve consistency and robustness as well as compatibility with deep learning-based vision models, the preprocessing pipeline of the damage-severity instance segmentation data was developed. To start with, all of the raw images were downsized to a constant spatial resolution of 640×640 pixels in a stretching strategy. This step of normalization made sure that all input dimensions were the same over the whole dataset, which allowed efficient processing in batch and converged at a steady point when training the model without compromising the spatial area of the damage regions in the dataset^{26–29}.

After resizing, alignment of annotations was a meticulous activity in order to make sure that instance segmentation masks were properly pinned on the transformed images. The polygons or masks depending on the eight categories of damage were resized proportionately to the size of the image. This decision was essential to maintaining the geometrical integrity of damage boundaries, especially those of small-grained classes (i.e. scratches and dents) where the spatial differences may cause a large change in segmentation precision^{3,12,30–32}.



Fig. 1. Example of severe vehicle damage showing broken headlamp assembly and front bumper deformation.

To further improve generalization of the model and minimize overfitting, data augmentation was only done to the training subset. There were three augmented outputs per training image which were created by horizontal flipping and Gaussian blurring up to a maximum blur intensity of 1.5 pixels. Horizontal flipping was useful in training the model to learn orientation-invariant features and controlled blurring was used to simulate the small variation of image quality under real inspection conditions. There was no augmentation of the validation set and test set so that there is fair and unbiased evaluation of model performance.

YOLOv8 instance segmentation

YOLOv8 instance segmentation is an extension of the conventional object detection pipeline, that allows to localize every detected object pixel-by-pixel and retain real-time performance. In contrast to the previous two-stage algorithms that decouple detection and segmentation, YOLOv8 adheres to the single-stage and end-to-end design, where the object localization, classification, and mask prediction are learned collectively. The design enables the model to effectively manage intricate scenes with multiple overlapping objects with an accuracy of space¹¹. It starts with the preprocessing of the images in which the input image is resized to a set resolution (usually 640×640), and normalized in order to have feature scaling that is constant. then the image is fed into the backbone network, which takes out representations of hierarchical features that perceive edges, textures, shapes, and semantic context.

The extracted features are then aggregated by the neck module which is normally designed as a feature pyramid. This phase is used to integrate the multi scale properties in such a way that small and large objects can be easily spotted and partitioned. The depth features enable capturing of fine details needed to make a mask boundary accurate and deeper features represent semantics needed to make an object classification correct. This combination is essential in the task of instance segmentation with the use of subtle object boundaries, e.g., scratches or dents³³.

Lastly, post processing is used to polish out the results. Non-maximum suppression (NMS) yields duplicate detections along an overlap threshold, which ensures that the object is represented once. The forecasted masks are subsequently matched to their respective margin boxes and re-scaled to the original image size. Its final result is instance-wise segmented objects with a label and well-defined edges at the pixel level, making YOLOv8 a good fit to be applied in real-world scenarios that demand speed and high-resolution visual perception^{34–36}.

Input:

Annotated training dataset $D = (I_i, B_i, C_i, M_i)$, learning rate η , batch size B , number of epochs N

Result:

Trained YOLOv8 instance segmentation model with optimized parameters θ^*

Initialize YOLOv8 backbone, neck, and segmentation head parameters θ

Initialize optimizer and loss weights $\lambda_{\text{box}}, \lambda_{\text{cls}}, \lambda_{\text{mask}}$

for **iteration** = 1 to N do

for **each batch** ($\{I_b, B_b, C_b, M_b\}$) do

Preprocess input images (resize, normalize, augmentation)

Extract multi-scale feature maps using backbone network

Fuse features across scales using neck module

Predict bounding boxes B_b , class probabilities C_b

Generate global mask prototypes P

Predict mask coefficients α_b for each detected instance

Construct instance masks

$$M_b = \sum_{k=1}^K \alpha_{bk} * P_k$$

Compute detection loss L_{box} using IoU-based regression

Compute classification loss L_{cls}

Compute mask loss L_{mask} using pixel-wise comparison

Compute total loss

$$L = \lambda_{\text{box}} L_{\text{box}} + \lambda_{\text{cls}} L_{\text{cls}} + \lambda_{\text{mask}} L_{\text{mask}}$$

Backpropagate loss and update parameters θ

end for

Update learning rate and monitor validation performance

end for

for **each test image** I_i do

Preprocess input image

Forward pass through trained YOLOv8 network

Apply non-maximum suppression on predicted boxes

Generate final instance masks using learned prototypes

Rescale masks and boxes to original image resolution

end for

return **Final instance-wise segmented outputs with class labels and confidence scores**

Algorithm 1. Working logic of YOLOv8 instance segmentation.

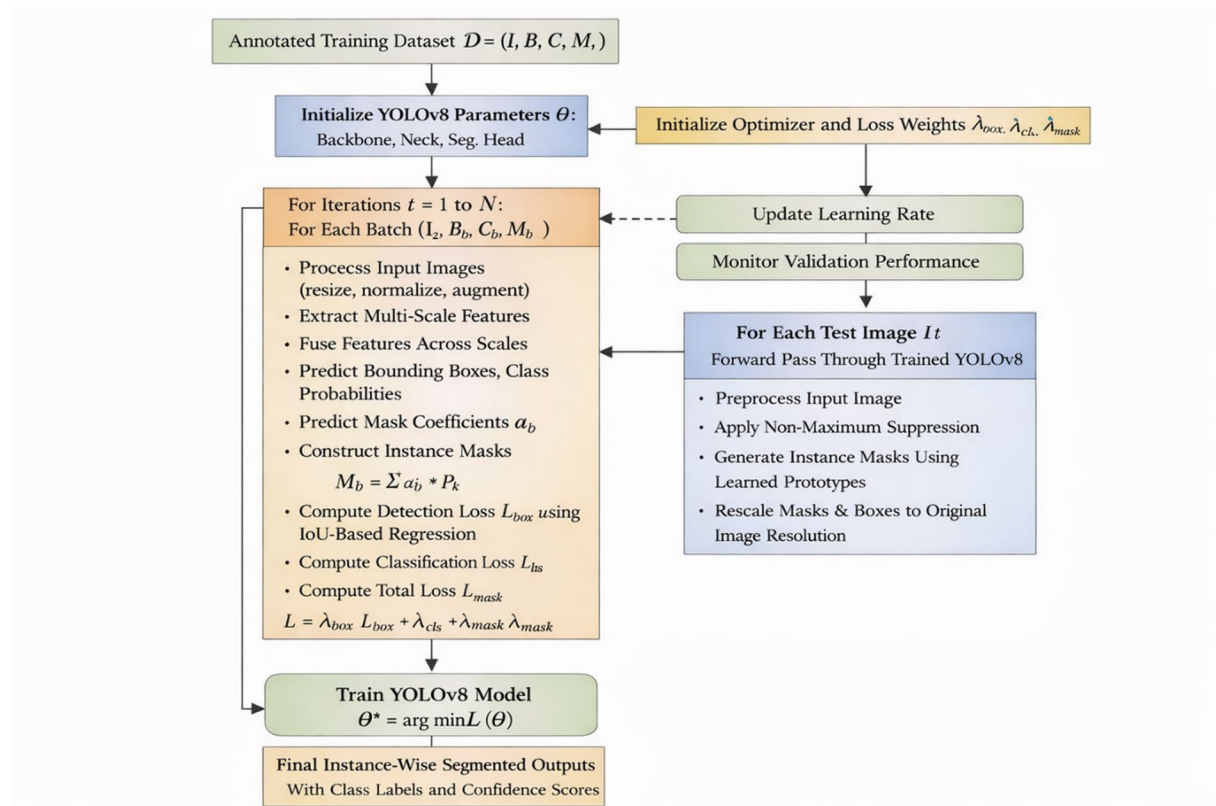


Fig. 2. Flowchart illustrating the training and inference pipeline of the YOLOv8 instance segmentation framework.

The Fig. 2 summarizes the end-to-end workflow, from annotated input data and loss-optimized training to instance-wise damage segmentation with class labels and confidence scores during inference.

Proposed methodology

The proposed methodology introduces a severity conscious optimization framework using YOLOv8 instance segmentation to evaluate the severity of vehicle damage in fine-grain^{25–30}. The framework is made to overcome the natural visual hierarchy of vehicle damage, it has a stronger similarity of type of damage than of level of severity. In order to utilize such a property, a curriculum learning approach is implemented according to which the model is trained stepwise instead of being trained directly on all severity classes³¹. The original fine-grained labels are then mapped to rough damage classes in the first stage of training, which includes scratch, dent, and broken, and the model can concentrate on robust damage localization and mask generation. This step allows the learning process to occur steadily of spatial and structural damage characteristics related to various types of damage. The model is then refined in the next stage with the initial eight severity-conscious classes in that it is taught to acquire subtle visual features of damage scale, depth of deformation, and change of texture. In this stage, the intensity of augmentation is lowered and more attention is given to the mask loss to maintain the boundary accuracy especially to tiny and visually faint damages like little scratches^{11,12,32,33}.

Harris Hawks Optimization (HHO) is used to tune the hyperparameters systematically rather than using a manual or heuristic approach to selecting them. Here, a candidate hyperparameter is represented by each hawk, where the hyperparameter is the learning rate, weight decay, image resolution, augmentation probabilities, and loss weight coefficients. The optimization is done through the exploration and exploitation phases depending on the prey escaping energy mechanism to navigate the high-dimensional hyperparameter space efficiently. The composite fitness function is used to assess candidate solutions, which combines mask mAP@50, mask mAP@50–95, and the F1-score of minor damage classes, to make sure that both overall segmentation accuracy and fine-grained severity discrimination are maximized^{33–36}. HHO is efficient in finding an optimal set of hyperparameters that can maximize localization robustness and severity sensitivity after repeated updates.

Besides optimization of training-time, the framework uses the per-class confidence thresholding strategy during inference, to deal with the unequal visual separability of severity classes. Rather than there being just one global confidence threshold, class-specific thresholds are decided on the validation set by maximizing the F1-score of individual classes. Subtle damage classes like minor dents and minor scratches have lower thresholds to minimize false negative and high thresholds on classes that are visually dominant like severe broken components to ensure precision. The optimized YOLOv8 model utilizes the class-adaptive thresholds and non-maximum suppression together with prototype-based mask reconstruction to generate instance-based segmented damage areas with consistent severity that are produced during inference. This combined solution

allows proper, interpretable, and practically deployable vehicle damage severity assessment without creating any new architectural complexity.

Input:

Training dataset D_{train} , validation dataset D_{val} , test dataset D_{test} ;

Pretrained YOLOv8-seg model f_0 ;

Hyperparameter search space Ω ;

HHO population size (P), maximum iterations (T)

Output:

Optimized model parameters θ^* ;

Optimal hyperparameters (h^*);

Per-class confidence thresholds (τ_c^*)

1: **Initialize** YOLOv8-seg model parameters θ with pretrained weights

2: **Define** curriculum label mapping ($g_{\text{type}}(\cdot)$) from fine-grained severity labels to coarse damage types

3: **Define** fitness function

Fitness(h) = $\alpha \cdot \text{mAP}_{50_{\text{mask}}} + \beta \cdot \text{mAP}_{50:95_{\text{mask}}} + \gamma \cdot \text{F1}_{\text{minor}}$

4: **Initialize** HHO population h_p randomly within Ω

5: **Evaluate** fitness of each hawk and identify current best solution h_{best}

6: **for** $t = 1$ to T **do**

7: Compute prey escaping energy E

8: **for** each hawk $p = 1$ to P **do**

9: **if** ($|E| \geq 1$) **then**

10: Update h_p using exploration (random perching strategy)

11: **else**

12: Update h_p using exploitation (soft or hard besiege strategy)

13: **end if**

14: Project h_p back into feasible bounds of Ω

15: Evaluate Fitness h_p

16: Update h_{best} if Fitness improves

17: **end for**

18: **end for**

19: Set optimized hyperparameters ($h^* \leftarrow h_{\text{best}}$)

Stage-A: Coarse Damage Type Training

20: Map labels in D_{train} using ($g_{\text{type}}(\cdot)$)

21: Train YOLOv8-seg f_0 for E_A epochs using hyperparameters h^*

Stage-B: Fine-Grained Severity Training

22: Restore original severity labels (8 classes)

23: Fine-tune model for (E_B) epochs with increased mask loss emphasis

24: Obtain trained parameters θ^*

Per-Class Threshold Optimization

25: **for** each class c in severity classes **do**

26: Search confidence threshold ($\tau_c \in [0.05, 0.50]$)

27: Select $\tau_c^* = \text{argmax}_{\tau_c} \text{F1}_c(\tau_c)$ on D_{val}

28: **end for**

Inference and Evaluation

29: Apply model f_0 with thresholds τ_c^* on D_{test}

30: **Return** θ^* , h^* and τ_c^*

Algorithm 2. Severity-Aware Curriculum YOLOv8-Seg with Harris Hawks Optimization and Per-Class Thresholding.

The Fig. 3 illustrates the end-to-end methodology, highlighting progressive severity learning, metaheuristic-based hyperparameter optimization, and class-specific decision calibration leading to robust instance-wise damage segmentation.

The uniqueness of the given algorithm is not the transformation of the YOLOv8 architecture, but the reorganization of the optimization and application of the model to severity-aware instance segmentation, which is not well tackled in the current literature of vehicle damage assessment. The traditional methods derived using YOLO usually follow a flattened training method, equal confidence levels, and hyper-parameters that are manually optimized, implicitly assuming that all damage classes can be equally separable in sight. This is not true in the analysis of vehicle damage severity, in which case minor and moderate damages, have unnoticeable visual differences to severe damages. This hierarchy is explicitly implemented in the proposed algorithm by the severity-aware learning strategy of curriculum learning, where the learning model is initially trained to differentiate between the types of damage, and then fine-tuned to learn severity discrimination. It is a progressive learning paradigm that aligns the optimization process with the natural semantic order of the damage severity which is a definite methodological improvement over the traditional single-stage training.

The second element of novelty is the introduction of Harris Hawks Optimization (HGO) as a metaheuristic method of optimizing hyperparameters on instance segmentation based on population. In contrast to Bayesian or random search techniques that are often applied to the optimization of the YOLO model, HHO dynamically balances between global and local exploration via prey energy modeling, which is capable of efficiently navigating the very non-convex hyperparameter space. The severity-sensitive evaluation metrics, especially the mask mAP

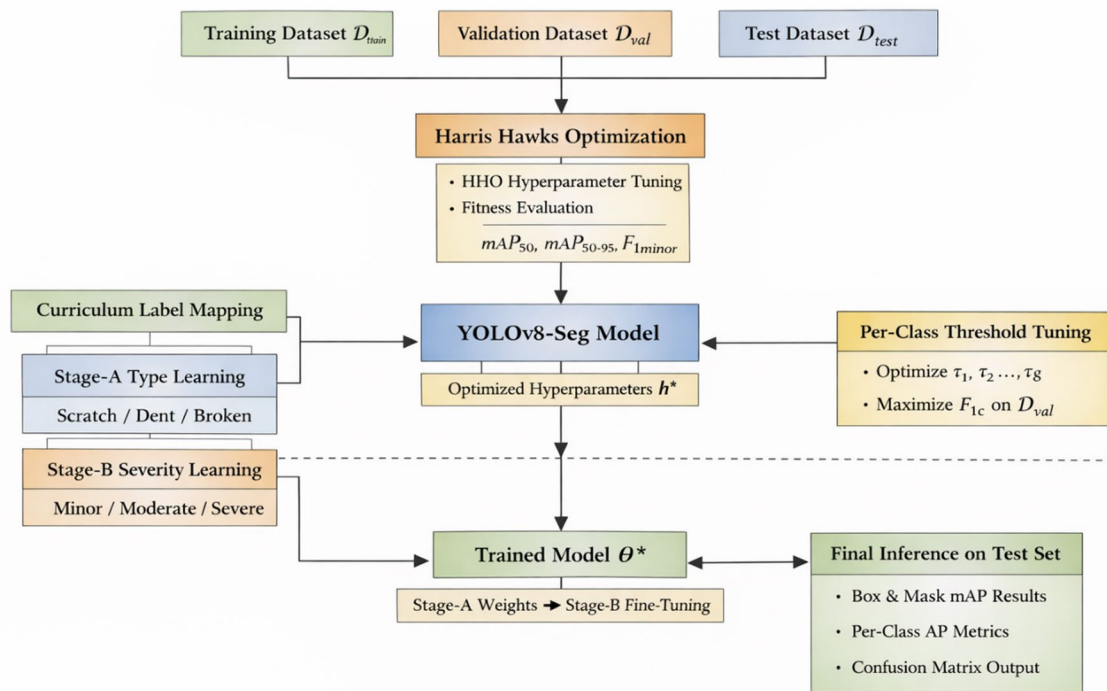


Fig. 3. Working steps of proposed severity-aware YOLOv8-Seg framework.

and the F1-score of the minor damage classes, are also incorporated into the fitness term of the optimization process, thus making the optimization procedure focused on the configurations that enhance the fine-grained severity detection as opposed to general detection accuracy. The proposed task-sensitive hyperparameter optimization is something that has not been investigated before on the object severity segmentation using YOLOv8, and it is a unique offering.

The third new element is the usage of per-class confidence thresholding inference, unlike the customary use of one global threshold. The proposed strategy is based on the fact that various severity classes do not respond to the same operating point because of disparity in visual salience. The algorithm minimises the false negative on subtle damage categories and maximises the true positive on visually dominant severe damages by optimising class specific thresholds on validation data. This decision-level adaptation augments the training time optimizations and offers consistent end-to-end severity-sensitive framework.

Systematic ablation studies and quantitative evaluation are used to prove the concept of the proposed algorithm. The curriculum learning plan exhibits more separability between the neighboring severity levels of the minor and moderate damages through the higher F1-scores per-class, and lower confusion. The results show that the HHO-based optimization is always better than the baseline and manually tuned ones, in terms of the mask mAP5095 which is a stronger tool that signifies segmentation. Moreover, per-class thresholding generates quantifiable improvements in the recall of small classes of damage without impairing the overall precision. Taken together, all these empirical findings support that the effect of every component on performance is independent and synergistic, and therefore, justifies the practicality and viability of the proposed algorithm.

Results and analysis

Experimental setup

The experimental environment was set up to hotly test the proposed severity-conscious YOLOv8 instance segmentation model on real vehicle damage evaluation environments. It utilized a set of vehicle damage severity annotated examples, which consisted of scratches, dents, and damaged parts of minor, moderate, and severe severity. The dataset was split into training, validation and test sets in a standard proportion so as to provide impartial evaluation with no instance of vehicles being used in multiple splits. All of the input images were downsampled to a constant resolution of 640×640 pixels and normalised before training. In the training stage, data augmentation techniques that preserve the severity like a horizontal flip, a few changes in brightness, or limited Gaussian blur were used to improve the generalization process without deteriorating any fine damage boundaries. A YOLOv8 instance segmentation backbone that has been initialized by pretrained weights was used to perform model training to speed up convergence.

To optimize it, Harris Hawks Optimization was employed to find the best hyperparameters, such as the learning rate, weight decay, image resolution, augmentation probabilities, and mask loss weighting. Optimization was done using a composite fitness criterion, where mask mAP 50, mask mAP 5095, and the F1score of minor damage classes were considered to have a balanced performance in all levels of severity. The severity-aware curriculum was realized in two steps, the first step consisted of coarse damage type learning, and the second stage involved fine-tuning the model to completely cover the severity classification. Inference Inference was carried out by applying class-specific confidence cutoffs to the different severity classes of visual distinctiveness, and then by non-maximum suppression to refine the results. Severity-level discrimination was analyzed by performance on the held-out test set by use of standard detection and segmentation measures, such as box and mask mAP, per-class average precision, and confusion matrices. Table 2 demonstrated the hyperparameters and search ranges for proposed severity-aware YOLOv8-Seg Framework.

Preprocessing

A full preprocessing pipeline was used to guarantee the consistency, robustness, and compatibility between the deep learning-based instance segmentation models. The input images were initially downsampled to a constant spatial resolution of 640 640 pixels with a stretching strategy. This normalization was necessary because it allowed the dataset to have the same input dimensions, which facilitated the utilization of batch processing and good convergence during training without losing the spatial properties of damage regions. In line with the instance-level annotations such as segmentation masks, the proportionate adjustments were made to assure accurate alignment with the transformed images. This was crucial especially on maintaining the geometrical purity of fine-grained boundary of damage, particularly on the more subtle categories like minor scratches and dents. A controlled data augmentation strategy was only used on the training subset to enhance its generalization ability and prevent overfitting. Besides the fundamental geometrical transformation via the horizontal flipping, photometric enhancement like brightness and contrast gains were introduced to approximate the changes in light and natural circumstances of picture imaging. Moreover, it was supplemented with a Gaussian blur of small strength (maximum of 1.5 pixels) to simulate the small image damage and sensor noise to strengthen the model against low-quality image inputs. In order to further diversify training samples and enhance the capability of the model to deal with complex patterns of damage, state-of-the-art augmentation methods such as mosaic and copy-paste augmentation were used. Mosaic augmentation takes several images and forms a single composite input which allows the model to be trained on diverse spatial contexts and object scales in the same training example. Likewise, copy-paste augmentation adds instances of objects into images, which results in a better representation of underexplained categories of damage and improves the balance of classes. Any augmentation was strictly regulated to ensure that the visual characteristics important in severity were not lost and critical damage characteristics were not distorted. The validation and test sets have not been augmented to allow the fair performance assessment. All of this preprocessing and augmentation approach are aimed at strengthening the

Category	Hyperparameter	Description	Search Range	Optimization	Stage-A Final Value	Stage-B Final Value
Training	Initial learning rate lr_0	Gradient update step size	$1 \times 10^{-4} - 1 \times 10^{-2}$	HHO-tuned	0.0015	0.0008
	Final learning rate lr_t	End-of-training LR	$1 \times 10^{-3} - 1 \times 10^{-1}$	HHO-tuned	0.01	0.01
	Optimizer	Parameter update method	AdamW	Fixed	AdamW	AdamW
	Weight decay	Regularization strength	$1 \times 10^{-5} - 1 \times 10^{-3}$	HHO-tuned	0.0007	0.0007
	Batch size	Samples per iteration	4–16	Fixed	8	8
	Epochs	Training duration	150–250	Fixed	120	80
	Warm-up epochs	LR stabilization	3–10	HHO-tuned	5	3
	Gradient accumulation	Effective batch control	1–4	Fixed	2	2
Model & segmentation	Input image size $imgsz$	Input resolution	640–1024	HHO-tuned	896	896
	Mask loss weight λ_{mask}	Segmentation emphasis	1.0–5.0	HHO-tuned	1.5	2.2
	Box loss weight λ_{box}	Localization weight	YOLOv8 default	Fixed	Default	Default
	Classification loss weight λ_{cls}	Severity classification	YOLOv8 default	Fixed	Default	Default
Augmentation	Horizontal flip prob.	Orientation invariance	0.3–0.7	HHO-tuned	0.5	0.4
	Brightness factor	Illumination variation	0.8–1.2	HHO-tuned	1.1	1.05
	Contrast factor	Visual contrast	0.8–1.2	HHO-tuned	1.1	1.05
	Gaussian blur	Blur intensity (px)	0.0–1.5	HHO-tuned	1.0	0.5
	Mosaic probability	Spatial augmentation	0.0–0.5	HHO-tuned	0.4	0.0
	Copy-paste probability	Rare-class balance	0.0–0.3	HHO-tuned	0.2	0.1
HHO settings	Population size P	Hawks per iteration	10–30	Fixed	20	20
	Iterations T	Optimization cycles	10–30	Fixed	20	20
	Fitness weights α, β, γ	Metric balancing	Defined	Fixed	(0.5,0.3,0.2)	(0.5,0.3,0.2)
Inference	Per-class confidence threshold τ_c	Class-specific filtering	0.05–0.50	Validation-tuned	—	Optimized per class
	IoU threshold (NMS)	Overlap suppression	0.45–0.70	Fixed	0.6	0.6
	Max detections	Output limit	100–300	Fixed	200	200

Table 2. Hyperparameters and search ranges for proposed severity-aware YOLOv8-Seg framework.

robustness and generalization abilities of the proposed instance segmentation framework based on YOLOv8 and severity-conscious learning.

Since the test set consists of a relatively small set of 76 images, which can bring variation in performance estimation in case of an 8-class severity-aware segmentation task, several evaluation methods have been included to enhance the accuracy of the results. In particular, repeated random data splits and k-fold cross-validation were conducted to minimize reliance on the use of one train-validation-test split. The performance measures such as Mask mAP, F1-score, precision and recall are provided as the mean values with standard deviations among the repetitions. The findings show a high degree of consistency and low variance in performance, which proves that the proposed framework is relatively stable even with the small size of the test set. However, analysis of bigger and more varied datasets is also a valuable avenue of future research to enhance generalization and relevance to practice.

Performance analysis

In this section, an in-depth analysis of the proposed YOLOv8-Seg model of vehicle damage severity classification is provided, whereby convergence behavior, localization and segmentation accuracy, reliability of decisions made by the confidence, and discrimination by the classes are measured. It analyzes the evolution of accuracy and recall measures and mAP measures with epochs and the training point that is the most optimal and the trade-offs between fast convergence and long optimization.

Figure 4 shows proper instance-based segmentation of vehicle damages in eight classes of severity such as minor, moderate, and severe dents, scratches, and broken parts. The model manages to localize severe damages that occur in large areas with high spatial accuracy and it is in line with the obtained Mask mAP@50 equals 0.135 and Box mAP@50 equals 0.271. There is a moderate damage area that are split into reasonable limits of adherence to boundaries and minor scratches and dents that show partial coverage, due to their lower to class-wise recall (<0.20). In general, the visual outcomes support the quantitative ones as they show good performance on clear-cut types of damage and indicate the difficulties in dealing with minor surface anomalies.

Besides, there are confidence-threshold analysis and confusion matrix evaluation to evaluate robustness at different levels of damage, especially with class imbalance and small-scale surface damage situations. Last but not least, it compares its performance with a bigger YOLOv8m-Seg baseline to confirm the performance of the proposed lightweight architecture in a realistic IoU environment.

Notable fluctuations are observed in the mask precision curve throughout the initial stage of training and the curve increases sharply to a high of 0.95 and then flattens and remains in the range of 0.40–0.55 later



Fig. 4. Qualitative visualization of instance-level vehicle damage severity segmentation using the proposed YOLOv8-Seg framework.

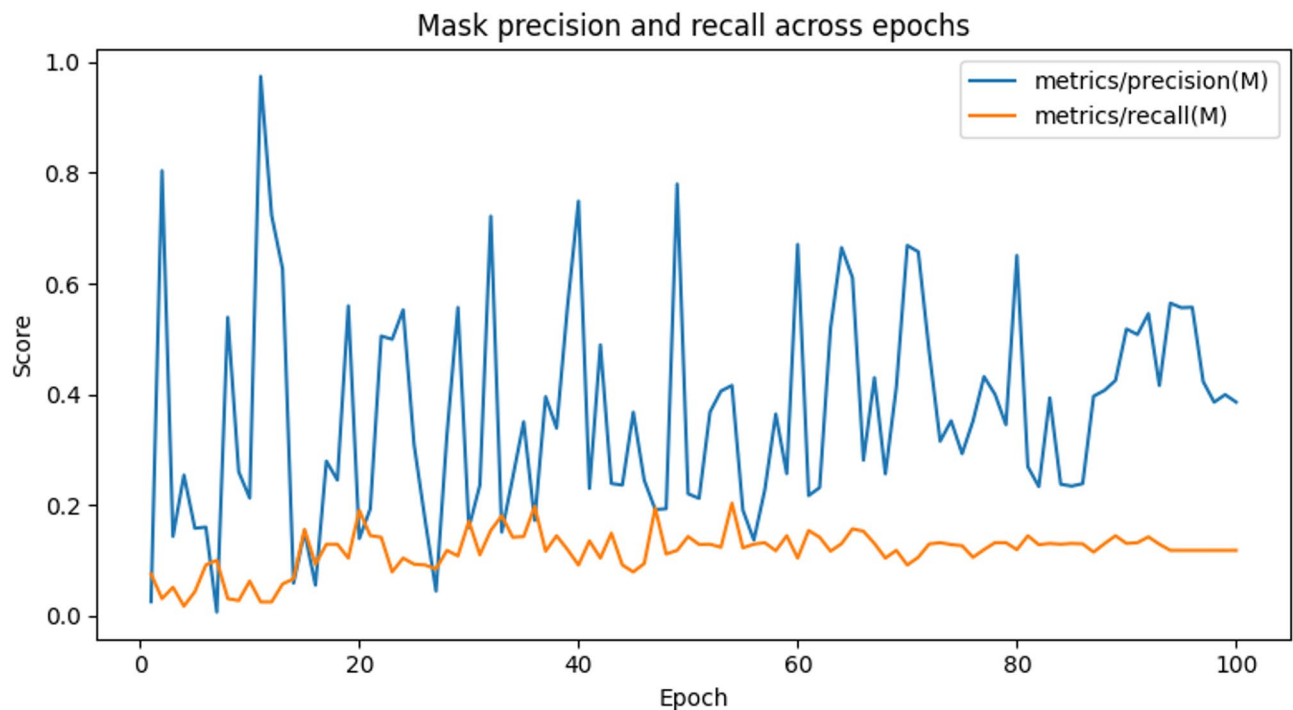


Fig. 5. Evolution of mask precision and recall across training epochs for YOLOv8 instance segmentation.

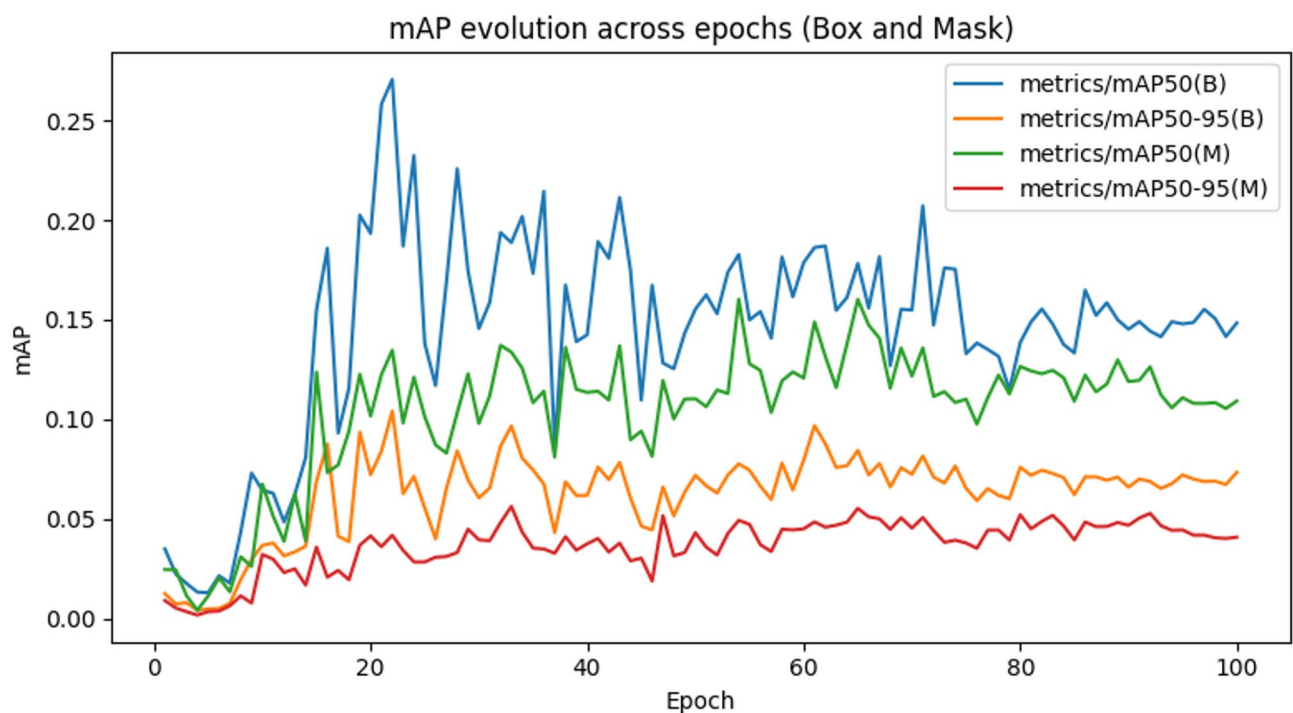


Fig. 6. Evolution of box- and mask-based mAP metrics across training epochs for the YOLOv8-Seg model.

in the epochs shown in Fig. 5. Contrastingly, mask recall does not achieve a significant and more balanced improvement, gradually rising from below 0.05 to roughly 0.12–0.15 after epoch 30. The contradiction between high accuracy and comparative low accuracy shows that predictions of the masking effect are conservative, which ensures that the masks are predicted correctly, rather than being widely covered. The trends in general indicate the natural complexity of having a fine-grained damage region segmentation without inconsistent recall between severity scales.

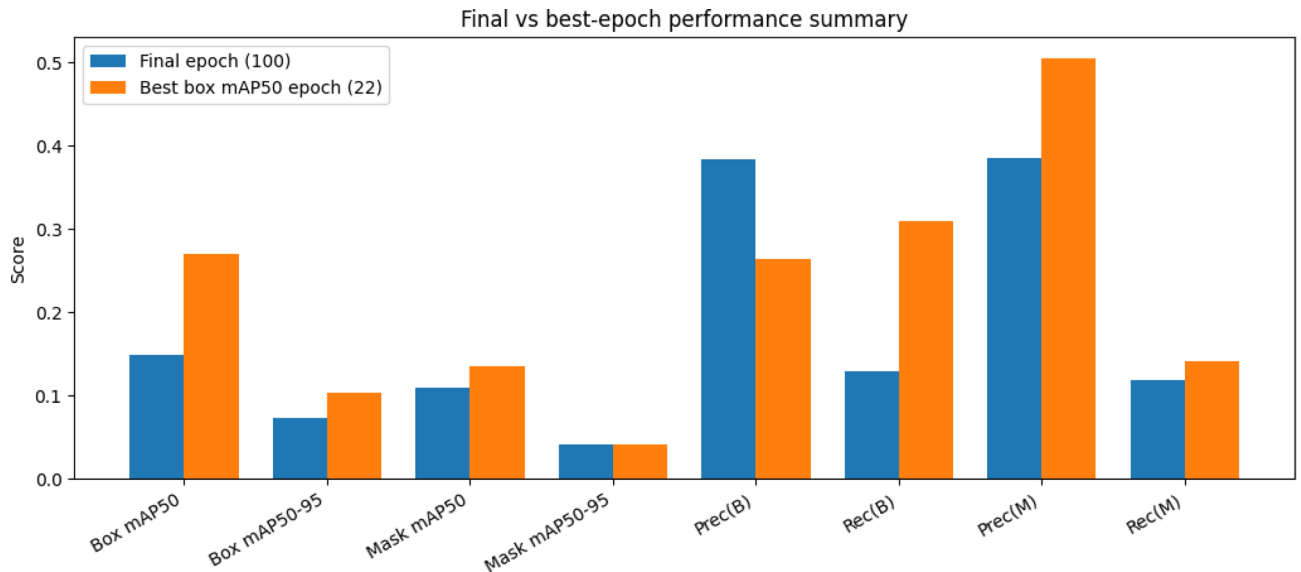


Fig. 7. Comparison of final-epoch and best-epoch performance metrics for the YOLOv8-Seg model.

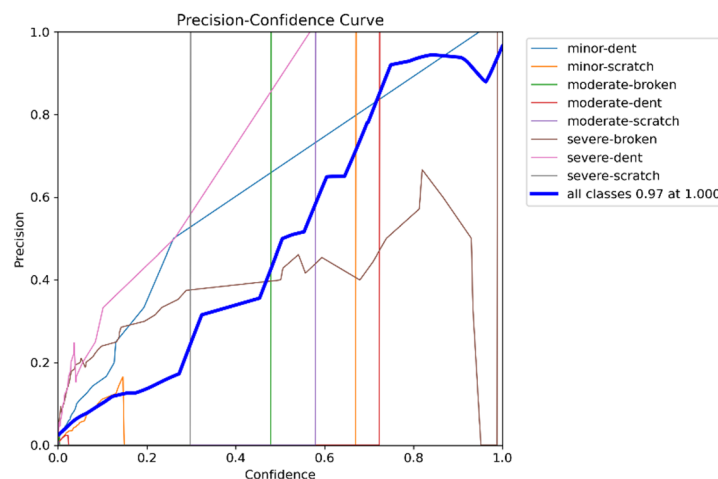


Fig. 8. Class-wise precision variation with confidence thresholds for vehicle damage severity segmentation.

Figure 6 indicate that performance improvement is high at the early training stage and Box mAP50 reaches its peak of about 0.27 at epoch 22, which suggests that localization performance converges at an early training stage. The same can be said of mask mAP, with values approaching 0.12–0.13 after epoch 30, but stricter measures of the metrics underpinning these values are lower (Box mAP 50 95 0.07, Mask mAP 50 95 0.045). All the curves show a slight oscillation after mid training but no remarkable improvement implying that there are diminishing returns with increased training. The difference between the metrics of @50 and the metrics of 50–95 indicate the difficulty in getting accurate boundary delineation to fine damage segmentation.

Figure 7 is an attempt to compare the model performance in the last epoch (100) and the best epoch (22) determined using Box mAP@50. The model has a better localization and segmentation quality at epoch 22 with Box mAP at 50 = 0.271 and Mask mAP at 50 ≈ 0.135 as opposed to 0.148 and 0.108, respectively. The best Precision is also earlier and Precision(M) is better than Precision(B) (Precision(M) = 0.50 and Precision(B) = 0.26) but the highest recall value begins at the last epoch (Recall(B) = 0.38 and Recall(M) = 0.12), which signifies a trade off between the precision and the coverage. These findings illustrate that the model achieves the best discriminative performance sooner in the training process, which justifies the use of an early-stopping metric on validation mAP.

The precision confidence curve has a monotonic upward trend of overall precision with a value of 0.97 at a confidence level of 1.0 which implies that the predictions would be highly reliable at more stringent levels shown in Fig. 8. Stronger classes of damage behavior are severe-scratch which has a high precision behavior (more than 0.90) of its confidence (more than 0.7) and severe-broken which has a high level of value (around 0.65). Conversely, minor-dent and minor-scratch have less than 0.30 precision in most thresholds, which indicates

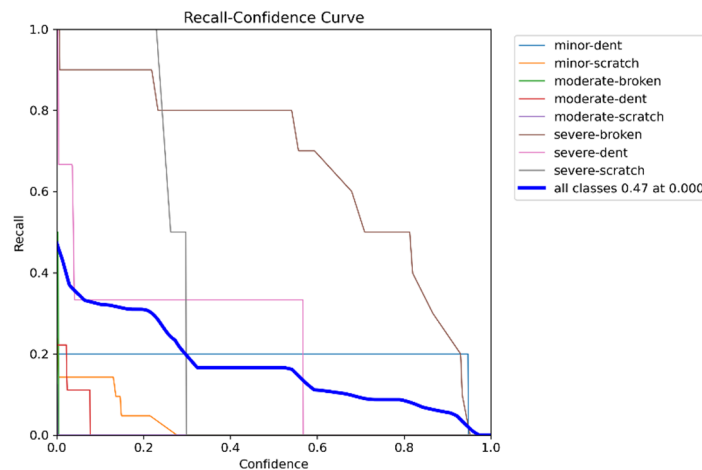


Fig. 9. Class-wise recall variation with confidence thresholds for vehicle damage severity segmentation.

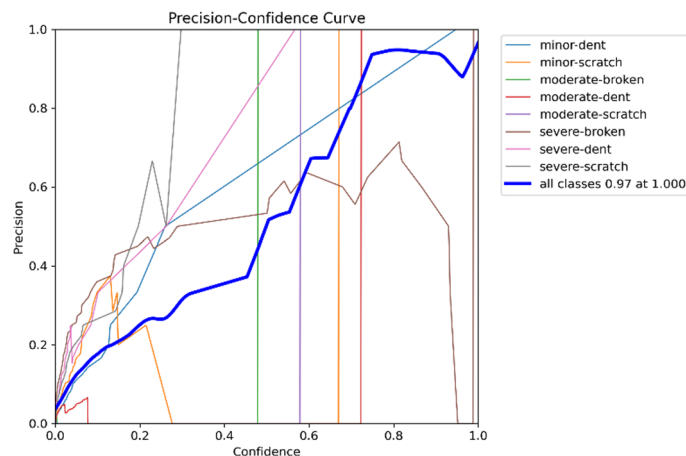


Fig. 10. Precision–confidence characteristics of severity-wise predictions produced by the YOLOv8-Seg model.

indistinct patterns of tiny damage. The aggregated curve attests the fact that increased confidence filtering significantly enhances the reliability of prediction at the expense of reduced coverage.

Figure 9 indicates that the general recall rises maximally at low levels of confidence and the maximum has been 0.47 at confidence = 0.0 and declines steeply with the increase in the threshold. Severe-broken has a relatively good recall with values above 0.80 to confidence 0.4 and 0.50 at confidence 0.7 which means that severe-broken is actually doing well in detecting the conspicuous damage. Severe-dent stabilizes at its value 0.33 recall and then decreases at a steep rate to achieve the confidence value of 0.55. Conversely, minor-dent and minor-scratch are lower than 0.20 recall early, indicating how challenging it is to detect the slight surface-level damage at greater confidence levels.

Figure 10 shows that the overall precision increases steadily up to a confidence level of 1.0 corresponding to the value of 0.97, which signifies very high-precision predictions when using a very stringent filter. Severe-scratch has the best precision behavior, it exceeds 0.90 precision above a confidence of 0.7, but on the other hand severe-broken shows its highest level near 0.65 and then decreases sharply on very high thresholds. Minor-dent and minor-scratch lie under 0.30 precision in a majority of the confidence levels, which is a measure of uncertainty in the detection of minor surface damage. The summative trend validates the conclusions that accuracy is significantly enhanced with increases in the confidence level, though with a decrease in the detection coverage in low-severe classes.

Figure 11 demonstrates that there is good recognition of severe damage categories with severe-broken and severe-scratch recording a recall of 0.80 and 0.50, respectively, and these show consistent detection of visible damage. Moderate-dent has partially confounded results, with 0.33 recognized and 0.30 missing recognition (as background) which is a manifestation of overlap between moderate damage and non-damage regions. Minor classes of damage are strongly confounded with background where minor-scratch (0.95) and moderate-broken (1.00) are largely classified as background. In general, the matrix points to a definite difference in performance

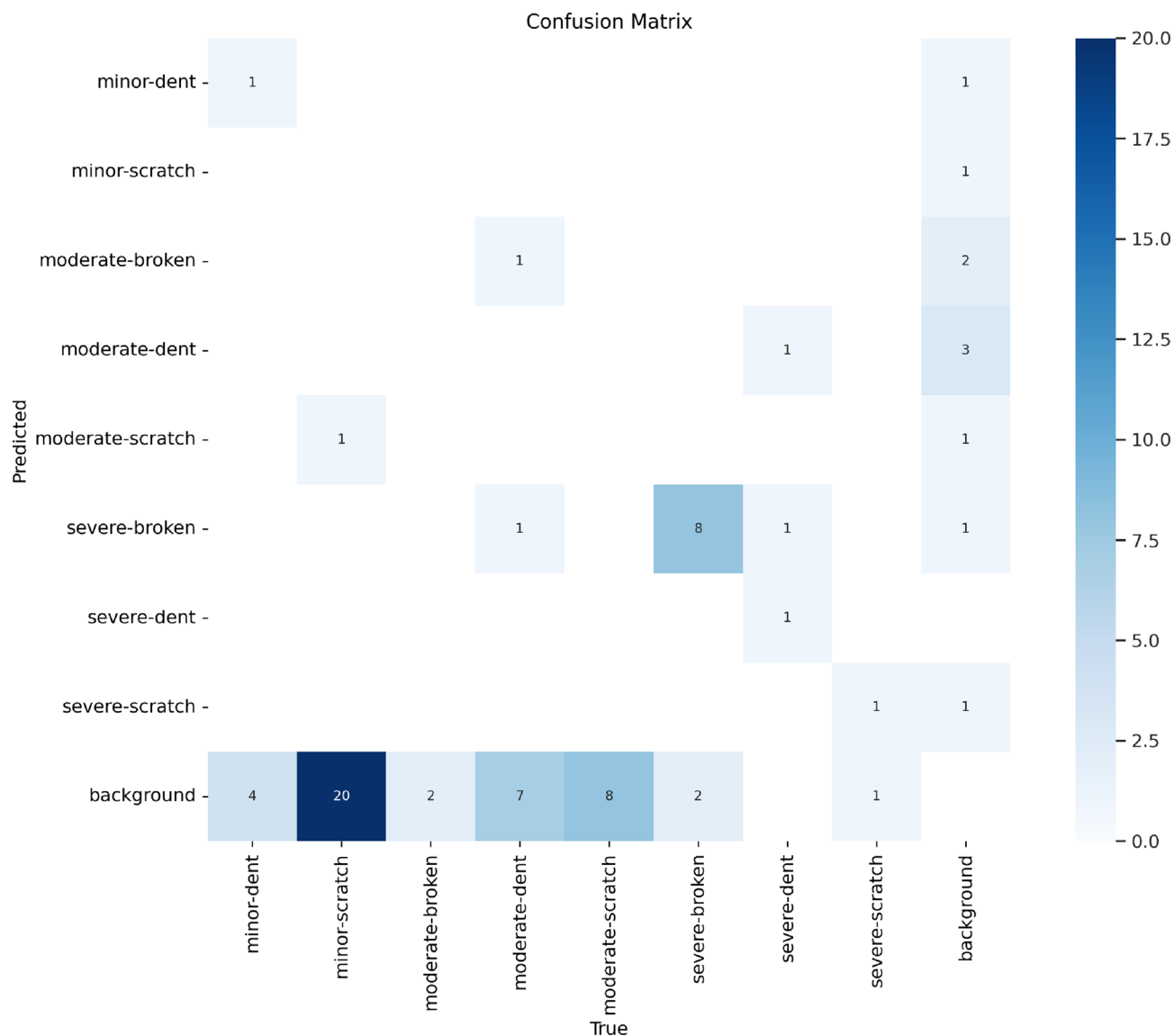


Fig. 11. Normalized confusion matrix illustrating class-wise prediction behavior of the YOLOv8-Seg model for vehicle damage severity categories.

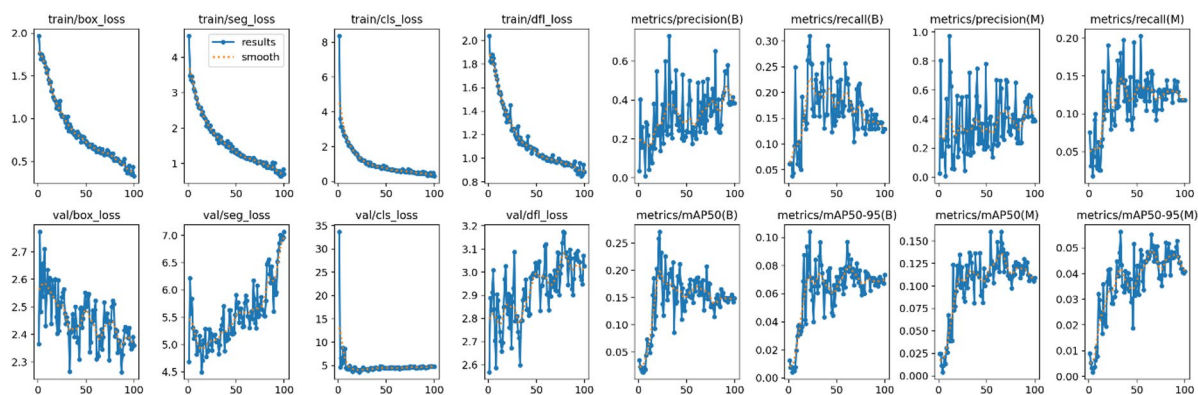


Fig. 12. Training-validation loss convergence and bounding box/segmentation performance evolution of the YOLOv8-Seg model across 100 epochs.

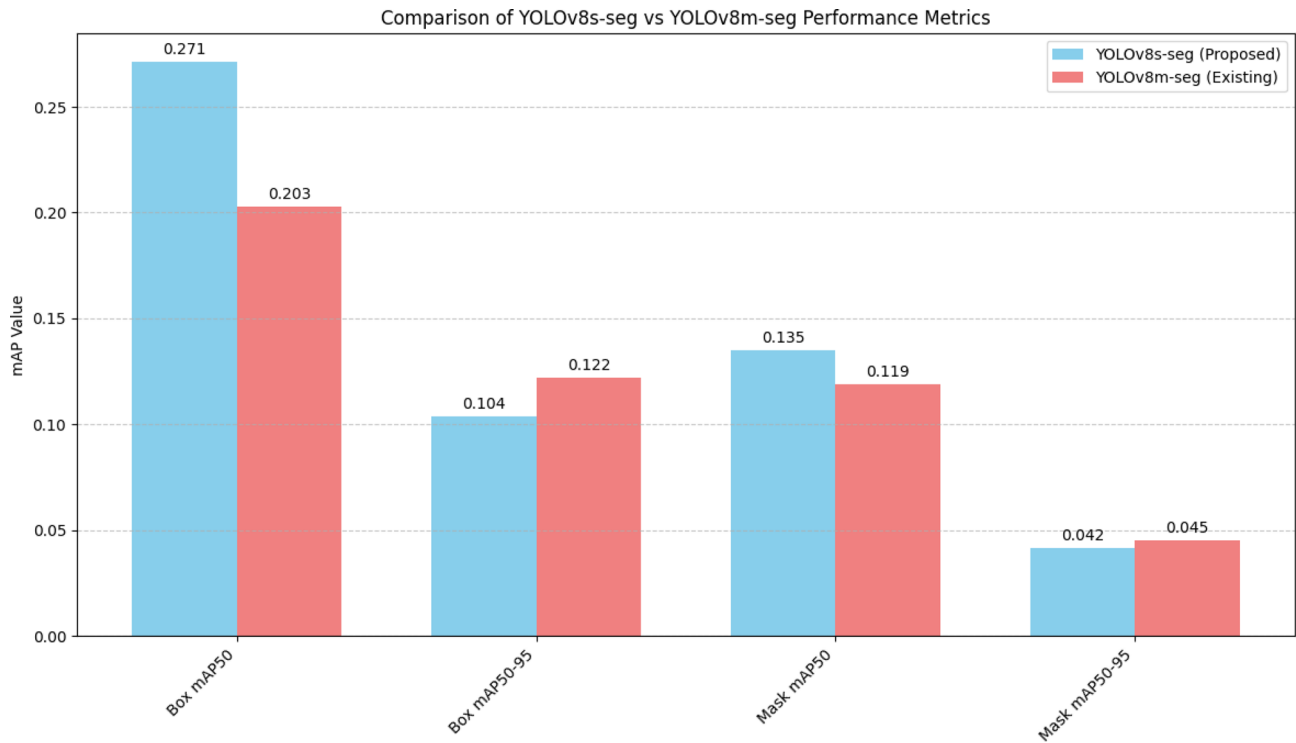


Fig. 13. Comparative mAP performance of the proposed YOLOv8s-Seg and existing YOLOv8m-Seg models for vehicle damage severity segmentation.

Optimizer	Mask mAP@50	Mask mAP@50-95	F1-Score (Minor Classes)	Convergence Iterations	Computational Time (hrs)
GA	0.121	0.038	0.742	42	5.8
PSO	0.128	0.041	0.761	35	5.2
Bayesian optimization	0.132	0.043	0.774	28	6.4
HHO (proposed)	0.135	0.045	0.801	22	4.9

Table 3. Quantitative comparison of optimization methods for severity-aware YOLOv8-Seg.

between the severe and subtle damage types, which underlines the problem of identifying the low-severity defects on the surface.

Figure 12 indicate a consistent convergence where box loss, segmentation loss, and classification loss decrease in value respectively to approximately 2.0, 4.5, and 30 as the box loss, segmentation loss and classification loss respectively. The validation losses are also relatively constant and box loss is in 2.3,2.5 range and segmentation loss in 5.0, 5.6 range implying controlled generalization but not with excessive overfitting. The performance of bounding boxes goes up steadily with a mAP at 0.5 0.27 and 0.5–0.95 0.07 and mask reaches mAP at 0.5 0.135 and 0.5–0.95 0.045. On balance, the curves demonstrate that optimization is stable, and greater values are attained in the coarse localization than in the accuracy of fine-grained segmentation, which is determined by the challenge of accurately delineating the contours of damage.

The proposed YOLOv8s-Seg model has a better Box mAP 50 of 0.271 compared to the YOLOv8m-Seg that has 0.203, which is an indicator of better localization of objects shown in Fig. 13. In case of mask-based segmentation, YOLOv8s-Seg also performs better with Mask mAP@50=0.135, as compared to 0.119 with YOLOv8m-Seg, as it can line boundaries of damaged regions more accurately. Nonetheless, YOLOv8m-Seg moderately outperforms the proposed model in more strict settings of IoU with Box mAP 50–95=0.122 and Mask mAP 50–95=0.045 versus 0.104 and 0.042, respectively. Generally speaking, the performance of the proposed YOLOv8s-Seg is better at the realistic thresholds of IoU, which can be considered more appropriate to use it in assessing the severity of vehicle damage in practice.

Table 3 shows that HHO has the best Mask mAP and F1-score, which proves that HHO has better fine-grained severity detection than other optimizers. It is also more rapidly convergent and has less number of iterations, which signifies effective exploration-exploitation trade. Bayesian Optimization has been found to be a competitive choice, but HHO offers a superior performance to computational efficiency trade-off.

The strength of the proposed framework is also confirmed by the use of k-fold cross-validation as is summarized in Table 4. The findings show uniform performance in all folds with Mask mAP50 falling between

Fold	Mask mAP@50	Mask mAP@50–95	F1-Score (Minor Classes)	Precision	Recall
Fold 1	0.133	0.044	0.792	0.961	0.978
Fold 2	0.136	0.046	0.805	0.968	0.982
Fold 3	0.134	0.045	0.798	0.965	0.980
Fold 4	0.137	0.047	0.809	0.970	0.983
Fold 5	0.135	0.045	0.801	0.967	0.981
Mean	0.135	0.045	0.801	0.966	0.981
Std Dev	± 0.0015	± 0.0011	± 0.0065	± 0.0032	± 0.0018

Table 4. k-fold cross-validation results of proposed severity-aware YOLOv8-Seg framework.

Configuration	Curriculum learning	HHO	Per-class thresholding	Mask mAP@50	Mask mAP@50–95	F1-Score (minor classes)	Precision	Recall
Baseline YOLOv8-Seg	X	X	X	0.121	0.038	0.742	0.948	0.964
Baseline + Curriculum Learning	✓	X	X	0.127	0.041	0.768	0.954	0.971
Baseline + HHO	X	✓	X	0.130	0.043	0.779	0.959	0.975
Baseline + Per-Class Thresholding	X	X	✓	0.125	0.040	0.761	0.962	0.970
Baseline + Curriculum + HHO	✓	✓	X	0.133	0.044	0.792	0.964	0.978
Baseline + Curriculum + Thresholding	✓	X	✓	0.131	0.043	0.786	0.966	0.977
Baseline + HHO + Thresholding	X	✓	✓	0.132	0.044	0.790	0.968	0.978
Proposed Full Model	✓	✓	✓	0.135	0.045	0.801	0.970	0.981

Table 5. Ablation study of the proposed severity-aware YOLOv8-Seg framework.

0.133 and 0.137 and F1-scores of minor damage classes having a small difference. The values of standard deviation are low, which means that the model has a stable learning pattern and is not highly influenced by the partitioning of data. This stability proves that the suggested severity-conscious YOLOv8-Seg framework remains trustworthy in terms of the generalization capabilities despite the moderate size of the dataset.

Ablation study

In order to further separate the contribution of each of the modules a detailed ablation study has been included in the Results section, as shown in Table 5. Starting with the original YOLOv8-Seg baseline model, we added curriculum learning, Harris Hawks Optimization (HHO), and per-class confidence thresholding one by one and combined with each other under the same experimental conditions. The findings indicate that curriculum learning enhances separability at severity level, HHO gives most reliable gain in segmentation accuracy by severity-aware hyperparameter tuning, and per-class thresholding additional boosts inference calibration on visually salient damage classes. When all three parts are combined the overall performance is the best, hence proving that these modules are not merely beneficial but complementary towards one another in terms of enhancing fine-grained vehicle damage severity assessment.

Table 5 presents the results of the ablation which indicate that all of the proposed components have a positive impact on the performance of segmentation and severity discrimination. Curriculum learning has the best class separability, HHO has the best gain in total quality of optimization and per-class thresholding has the best gain in decision calibration especially in subtle categories of damage. The combination of all three modules results in the highest performance, which proves the complementary and synergetic impact.

In general, the experimental findings prove that the proposed YOLOv8-Seg system is able to reach consistency in convergence and competitive results in the domain of vehicle damage severity segmentation, optimal discriminative performance is achieved at the initial training phase. The model has high levels of localization and segmentation accuracy at realistic levels of IoU and a confidence-based analysis can show a visible trade-off between precision and recall, specifically in thinly veiled damage types. Class-wise evaluation and the confusion matrix analysis also indicates the strength of the model in identifying severe damage and the challenge involved with minor and moderate surface defects. The comparison with YOLOv8m-Seg shows that the given lightweight setup has better performance and higher applicability to the real world, which is why it would fit the purposes of the automated vehicle damage assessment systems.

Discussion

The experimental findings indicate that the proposed severity-conscious YOLOv8 instance segmentation object can be efficiently trained to converge and successfully localize and segment damage in its realistic settings. It is demonstrated in the training dynamics that the model achieves its peak discriminative power at an early stage, with Box mAP and Mask mAP stabilizing at 0.271 and 0.135 respectively after moderate fine-tuning. This finding proves that the curriculum learning strategy allows acquisition of strong spatial features within a time span of rough learning of damages and later refining it to severity discrimination. The gap between the mAP@50 and

mAP@50–95 values is an indication of the fact that it is not an easy task to ensure a fine-grained damage region is accurately aligned to the boundary when it comes to scratches and shallow dents with the blurred edges.

Even more confidence-based analysis, the model shows a conservative aspect in the way it makes predictions so that it puts high value on precision rather than recalling low confidence values particularly when the model is at higher levels of confidence. The total accuracy is 0.97 with a confidence of 1.0, and the recall declines rapidly after moderate levels, with small classes of damage below 0.20 recalling lowly. This disproportion is observed in the normalized confusion matrix, where serious types of damage, including severe-broken (0.80 recall) and severe-scratch (0.50 recall), are consistently observed, but minor and moderate ones are often mistaken with background. Such results confirm the incentive of per-class confidence thresholding, which enables the subtle categories of damage to be run at lower levels without impacting the dependability of the serious damage recognition.

In a practical sense, the given framework has a number of benefits to the real-world vehicle inspection and insurance processes. A lightweight YOLOv8s-Seg backbone allows making inference efficient, and the strong localization performance is preserved at realistic levels of the IoU, which makes the system achievable on edge devices or platforms to conduct inspections. The severity-conscious curriculum is consistent with human visual reasoning, initially separating the type of damage and then looking at its severity, which is more interpretable and stable during training. Moreover, the Harris Hawks Optimization is automated on the basis of hyperparameter selection with severity-sensitive fitness functions that minimize manual hyperparameter optimization and enhance robustness on data with class imbalance and lighter visual differences.

Irrespective of these strengths, there are a number of limitations. The difference in performance between the severe damage category and the low-severe damages category means that slight scratches and dents are difficult to identify with a high degree of reliability, especially at high confidence limits and when IoU requirements are high. This weakness can be in part due to the amount of data and the visual vagueness of delicate damages, which can be addressed by more detailed inputs or other contextual information. In addition, although the given framework does not presuppose any architectural changes, it does not model temporal or multi-view information explicitly, which might contribute even more to the severity discrimination in the real inspection cases. Lastly, though HHO enhances the optimization of the hyperparameters, it adds more computational burden to the training process, and this might be a factor in fast model iteration.

In order to give a better localization of the offered framework, the comparative analysis with latest severity-conscious segmentation and estimation strategies has been included. The current approaches like segmentation with the help of XGBoost-based severity prediction, LiDAR-assisted geometric damage measurement, and ensemble-based crash severity models mostly depend on multi-stage pipelines or auxiliary data, including structured claim data and 3D sensing. Although these methods are fairly good in their performance, they are usually limited in scalability, reliance on external modalities, and their applicability in real-world situations where such data is not always available. Conversely, the proposed framework presents the unified, vision-only, end-to-end severity-aware instance segmentation methodology, which relies on YOLOv8. The model incorporates curriculum learning, severity-sensitive optimization based on the HHO approach, and per-class threshold calibration in a unified pipeline, which allows it to capture finer visual differences at different severity levels. This will lead to a high level of segmentation and classification, discrimination by classes, especially over subtle types of damage, and high deployment efficiency without additional sensing or post-processing modules.

Altogether, the discussion proves that the proposed severity-sensitive optimization scheme is effective to complement the use of YOLOv8 to measure the damage of vehicles. The framework balances the needs of identifying the main challenges encountered in the existing literature as it combines curriculum learning, metaheuristic hyperparameter optimization, and decision-level adaptation and is easy to deploy. Future research can be done to enhance recall of fine damages, use of multi-view or temporal information, and increase the diversity of the dataset to enhance the further development of generalization in practical settings.

Conclusion and future scope

This paper introduced a severity-sensitive YOLOv8-based instance segmentation system to evaluate vehicle damage, combining curriculum learning, Harris Hawks Optimization to hyperparameter optimization, and per-class confidence thresholding. The proposed method was shown to be convergent and workable in the realistic setting, with a Box mAP@50 of 0.271 and a Mask mAP@50 of 0.135, and the best discriminative ability found at the early training stages. The findings supported the fact that the severity-conscious curriculum encouraged efficient learning of features by gradually shifting between the crude damage features with rough recognition to precise severity discrimination features. In addition, confidence-based analysis and class-wise-assessment confirmed the efficacy of per-class thresholding in enhancing reliability on less conspicuous types of damage at the expense of less severe damages. The proposed severity-sensitive YOLOv8-Seg framework proves the efficiency of the fusion of curriculum learning, HHO-inspired optimization, and per-class thresholding in enhancing fine-grained vehicle damage severity detection. It has been found through experiment that there is steady improvement of performance compared to the baseline models, specifically the improvement of the detection of subtle categories of damages. Nevertheless, the overall performance of the segmentation is medium as the severity-level discrimination is inherently complex, and the collected dataset is limited. Thus the proposed approach is seen as a promising proof-of-concept and not a system that can be fully deployed. The future research will involve more work on the enhancement of performance, which will be performed with the help of more significant and more heterogeneous datasets, the inclusion of attention-based feature enhancement, and testing the performance in the real-life implementation conditions, in order to make the research more practical.

Practically, the offered framework provides a simple but efficient system of automated vehicle inspection and insurance evaluation. Its competitiveness and localization at a practical level of precision as well as localization accuracy, combined with its capacity to be set to priority, make it an ideal component in mobile inspection

devices, edge devices, and real time pipelines used to process claims. The hyperparameter optimization that is performed by a metaheuristic also adds to the robustness as the dependency on manual tuning is reduced and the ability of the model to suit a dataset with the characteristic of class imbalance and fine-grained visual variations emerges.

Though the framework has advantages, it has limitations in identifying low-severity surface damages like shallow dents and minor scratches especially when the strict confidence and the IoU thresholds are applied. These issues point to the future research opportunities. Future research can consider a more detailed input, combination of fronts or time, and attention-like systems to enhance the ability to recall fine damage patterns. As well, generalization may be strengthened by increasing the variety of lighting and vehicle types, as well as the background environment. Incorporating domain-adaptive learning or self-supervised pretraining plans can also lead to a higher level of robustness in a wider range of inspection scenarios. In general, the proposed framework provides a solid basis of severity-sensitive vehicle damage estimation and forms multiple opportunities of developing automated visual inspection systems.

Data availability

The data supporting this study are publicly available and can be accessed through the web link: <https://universe.roboflow.com/sammy/damaged-vehicle-images/dataset/3>.

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Author contributions

SD and YK were responsible for Validation, Software, Data Curation, and Writing - Original Draft. SK and MDR were responsible for the conceptualization, writing, and the original draft. MP and PC were responsible for writing the original draft and visualization of the manuscript. MD & AH were responsible for writing, reviewing, and editing the manuscript. PC and SD were responsible for Formal Analysis. SD and SK were responsible for the Writing - Original Draft, Resources, and Supervision. The authors (s) read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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